

GPT and Attention Mechanism

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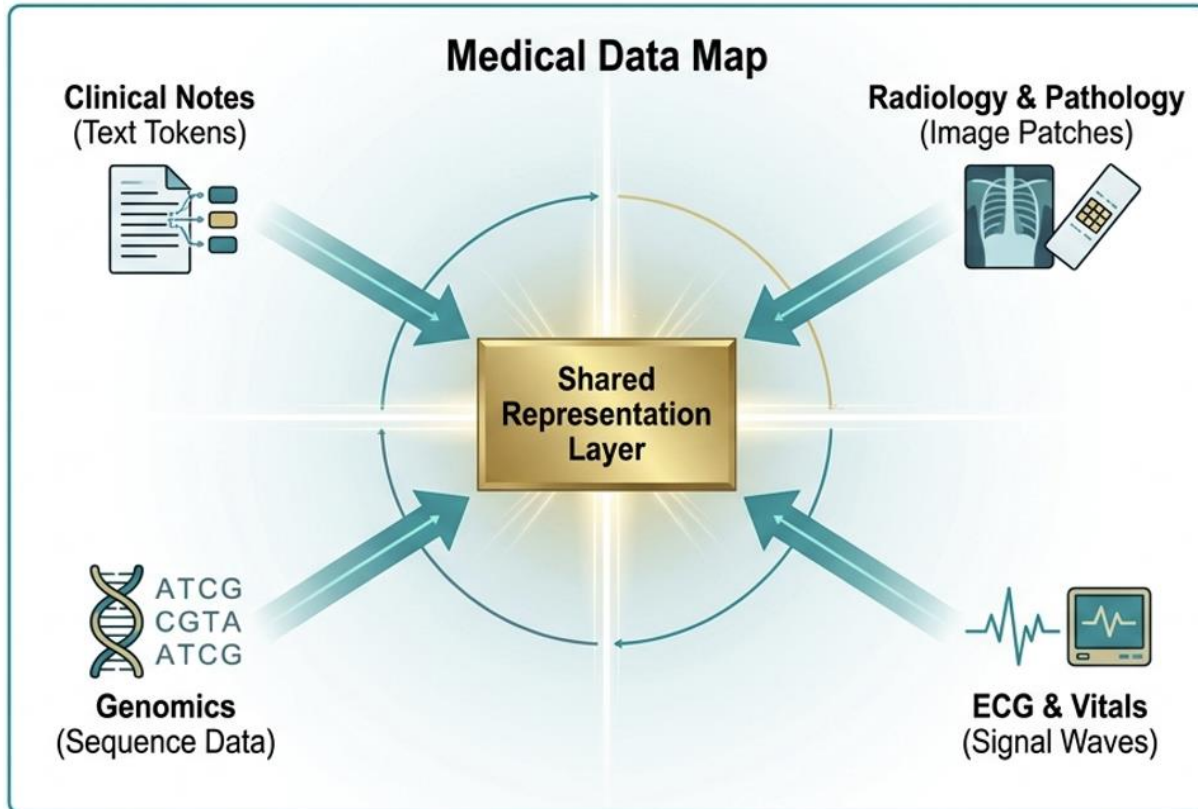
Medical Data is inherently multimodal, but attention is the foundation

The Challenge:

- Clinical decision-making integrates images, signals, and structure data, yet LLMs natively operate on order text tokens.

The Solution:

- Clinical decision-making integrates images, signals, and structure data, yet LLMs natively operate on order text tokens.

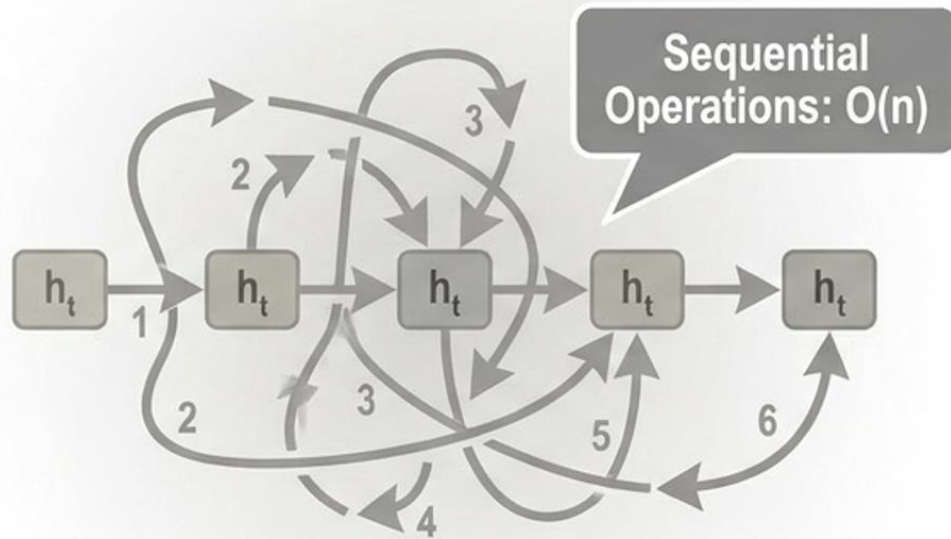


Papers to be discussed

- Vaswani A, et al. **Attention is all you need**. In: Advances in Neural Information Processing Systems (NIPS 2017); 2017. -- Introduced the **Transformer** architecture, a sequence model based entirely on attention mechanisms that outperforms RNN/CNN-based architectures while being faster to train
- Brown TB, et al. **Language models are few-shot learners**. In: Advances in Neural Information Processing Systems (NeurIPS 2020); 2020. -- Showed that GPT-3 (175B parameters) can perform a wide range of NLP tasks with only a few in-context examples, without any gradient updates or fine-tuning – the start of prompt engineering
- Singhal K, et al. **Large language models encode clinical knowledge**. Nature. 2023. -
- Demonstrated that instruction-tuning (with Med-PaLM) achieves state-of-the-art performance on medical QA benchmarks, while also revealing key gaps compared to clinicians through human evaluation – medical knowledge acquisition by QA with LLMs

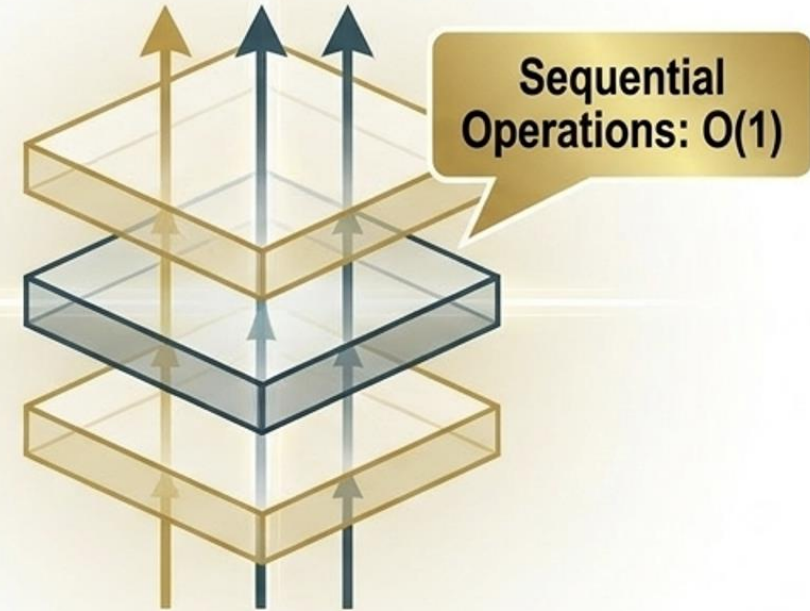
The Transformer Revolution

Past - RNN/CNN



Previous models factored computation sequentially, preventing parallelization.

Present - **Transformer** and Deep Slate

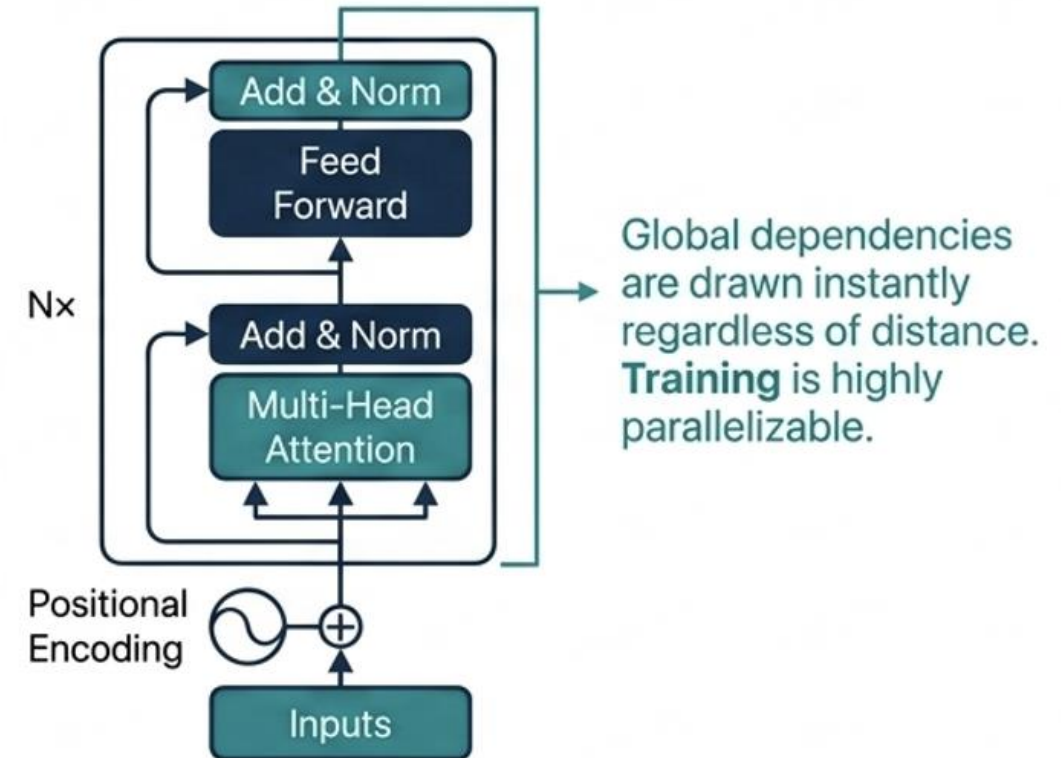
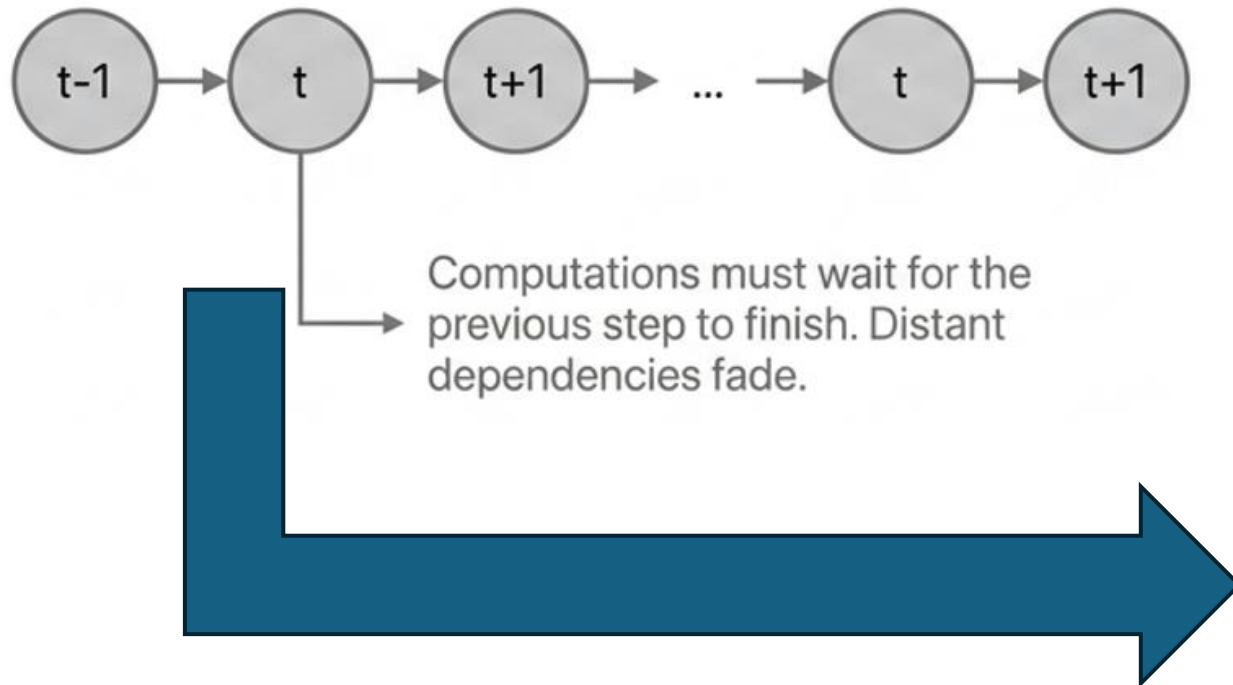


Dispenses with recurrence entirely. Parallelization enables massive scaling and solves the long-range dependency problem.

- RNNs struggle with long-range dependencies and cannot be parallelized due to sequential computation
- CNNs require many layers to capture distant relationships
- Transformer resolves both by using pure self-attention to model all pairwise token relationships simultaneously in each layer

Parallelize sequence information

The sequential bottle net



Remove recurrence and
unlock parallel processing

Attention as matching mechanism for tokens

"The unbelievable cat sat quietly"

tokenize



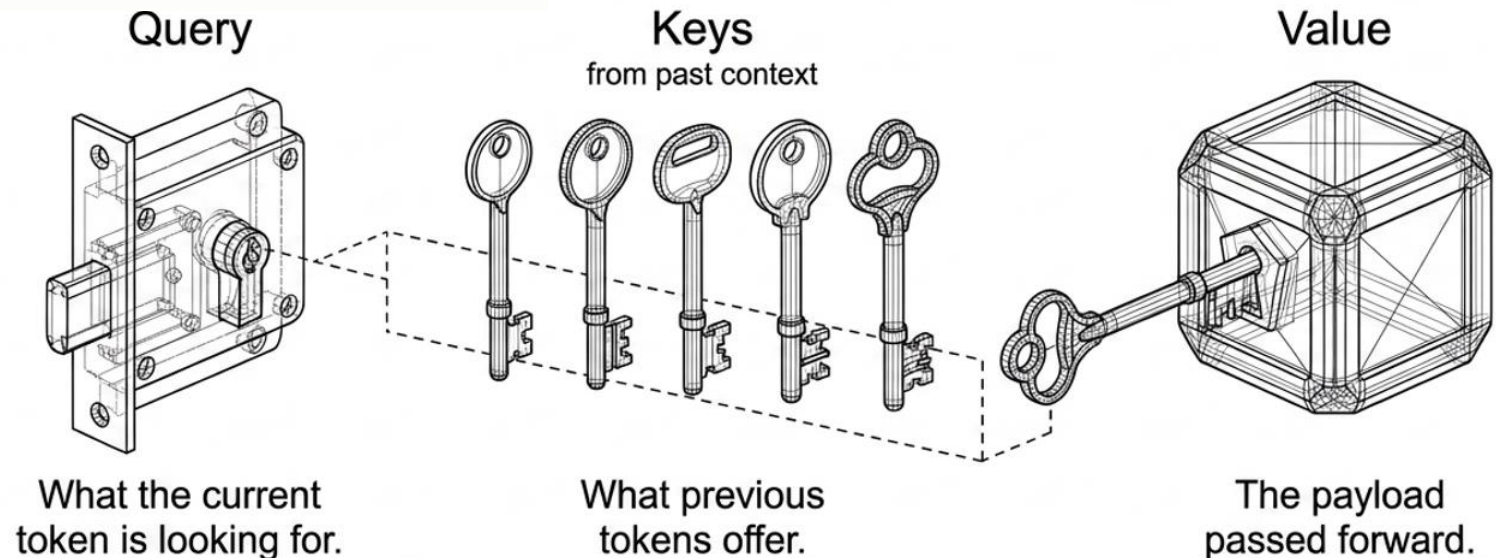
one word → 3 tokens

Each box = 1 token · Purple = common word · Teal = subword pieces · Coral/Amber = single tokens

Token as basic text unit
Chunk of text by sub-word algorithms such as BPE

Attention lets each token "look at" every other token and decide how much to borrow from each one to build its contextual meaning.

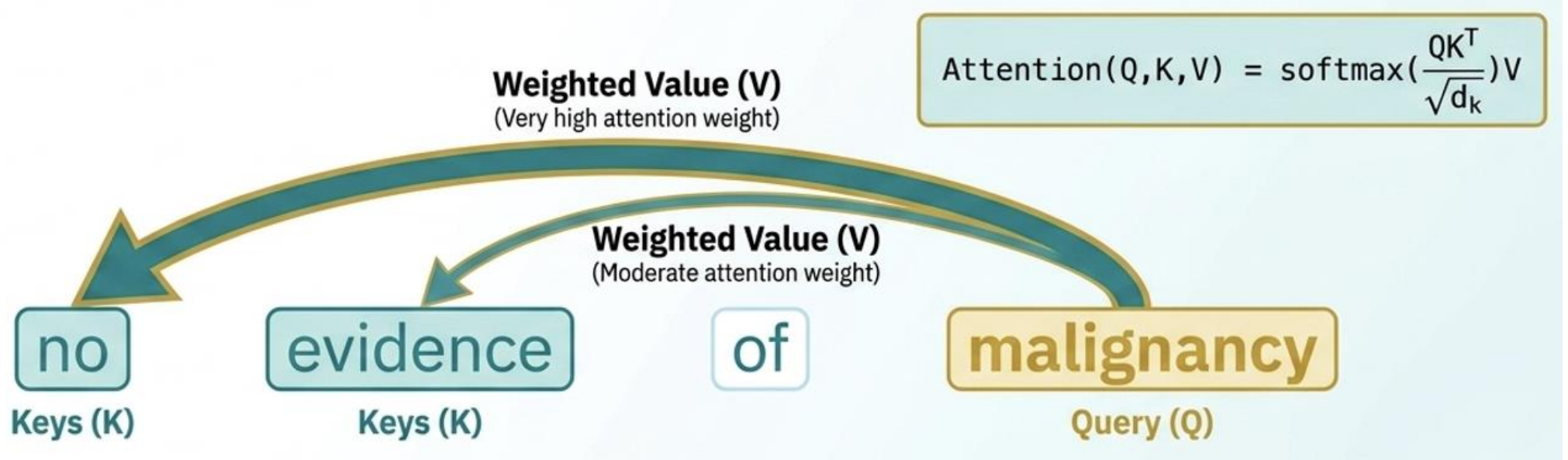
- **Query:** What am I looking for?
- **Key:** What do I represent?
- **Value:** What information do I carry?



The Math: $\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k})V$

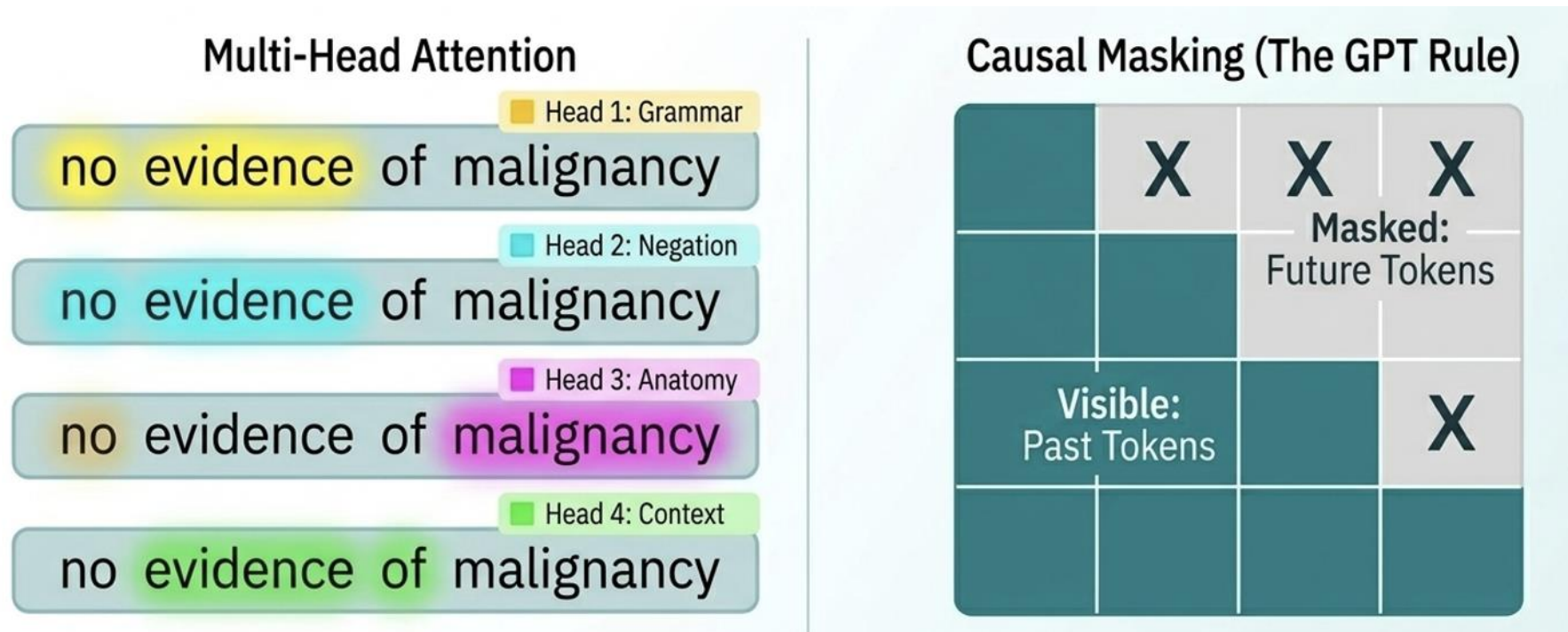
Computing attention scores

Attention weights each token in context simultaneously



- Query token “malignancy” ask who (K) and what (V) is relevant to me?
- Every token answers with a relevance score (K) and a meaning payload (V)
- Scaling factor ($\sqrt{d_k}$) prevents large dot product values from pushing the softmax function into a small gradient region (gradient vanishing)

Multi-head attention and causal masking



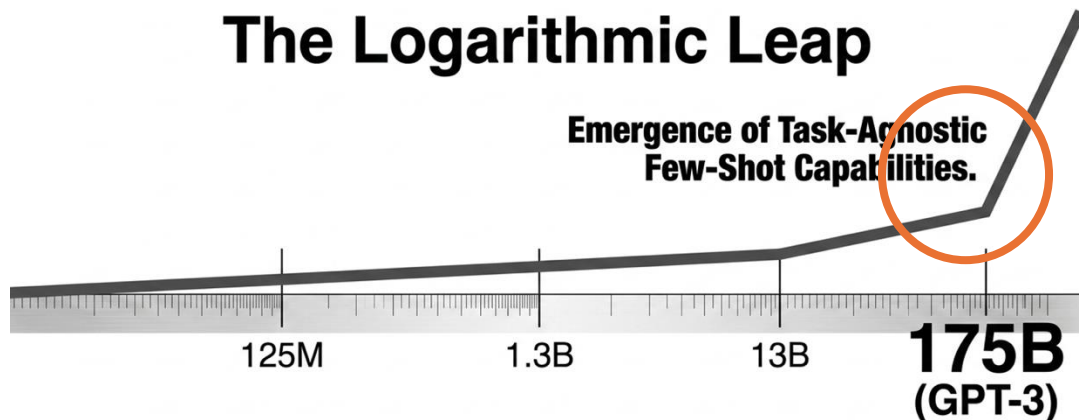
- The Q-K-V attention process runs in parallel multiple heads; thus, the model learns different subspaces
- The module is stacked many times and each token's representation becomes progressively richer – encoding long range knowledge
- Causal masking prevents cheating using a lower triangular matrix to set all future position to $-\infty$ before the softmax function

GPT-3 and task-agnostic few-shot learning

- LLM are Few-Shot Learners: GPT-3 as 175B-parameter, autoregressive LLM is 10x larger than any prior models
- In-context learning: task-specified learning using prompts alone, no backward gradients, no fine-tuning
- Taxonomy: zero-shot / one-shot / few-shot, as a rigorous evaluation framework
- few-shot performance scales predictably with model size, not architecture?
- Challenged the pre-train → fine-tune paradigm for foundation models
- Laid the conceptual groundwork for InstructGPT, RLHF, chain-of-thought prompting, and RAG
- Reshape the meaning of “learn” of AI – still an active topic

Taxonomy of In-Context Learning

	Zero-shot	One-shot	Few-shot
Examples in prompt	0	1	A few (2-20 +)
Weight update	No	No	No
Where learning happens	Pre-training	In context	In context
Task communicated by	Instruction only	1 example	N examples
Token cost	Minimal	Low	Moderate to high
Reliability	Low	Moderate	High



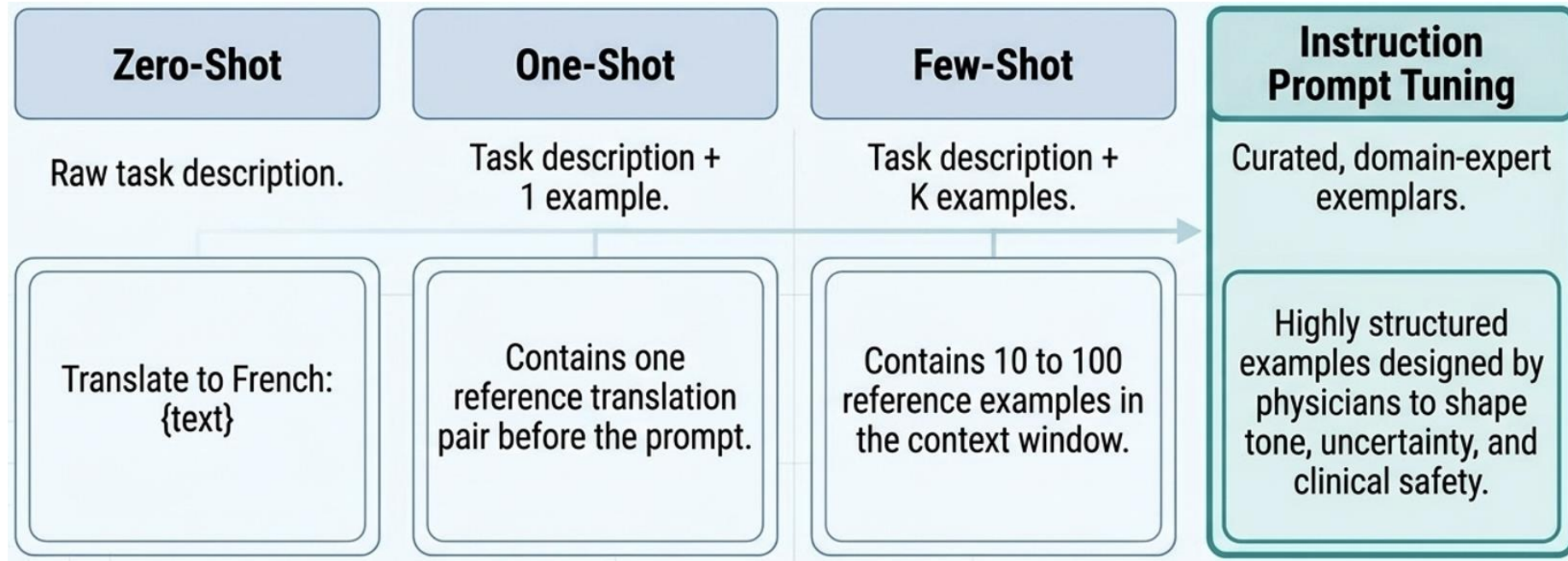
Performance improves gradually, and the few-shot learning starts to “launch” at a certain point (175B)

In-context learning becomes the foundation of LLM training

- In-context learning (ICL)
- Prompting engineering
- Chain of thought (COT)
- Instruction tuning / Reinforcement learning with human feedback (RLHF)
- Retrieval augmentation generation (RAG)
- Agentic prompting
- ? Distilled learning – No

ICL and few-shot learning in medical AI

Prompting from raw prediction to domain aligned output



Passing USMLE does not automatically become a good doctor.

- Few-shot learning reduces the need for large dataset
- Singhal et al asked the safety concerns when factual errors can harm patients
- MultiMedQA benchmark: combination of exam style QA
- HealthSearchQA: a new dataset to align QA from exam style to clinical style
- Med-PaLM: demonstrate Instruction tuning help unlock latent knowledge encoded in LLM, but remains inferior to clinicians

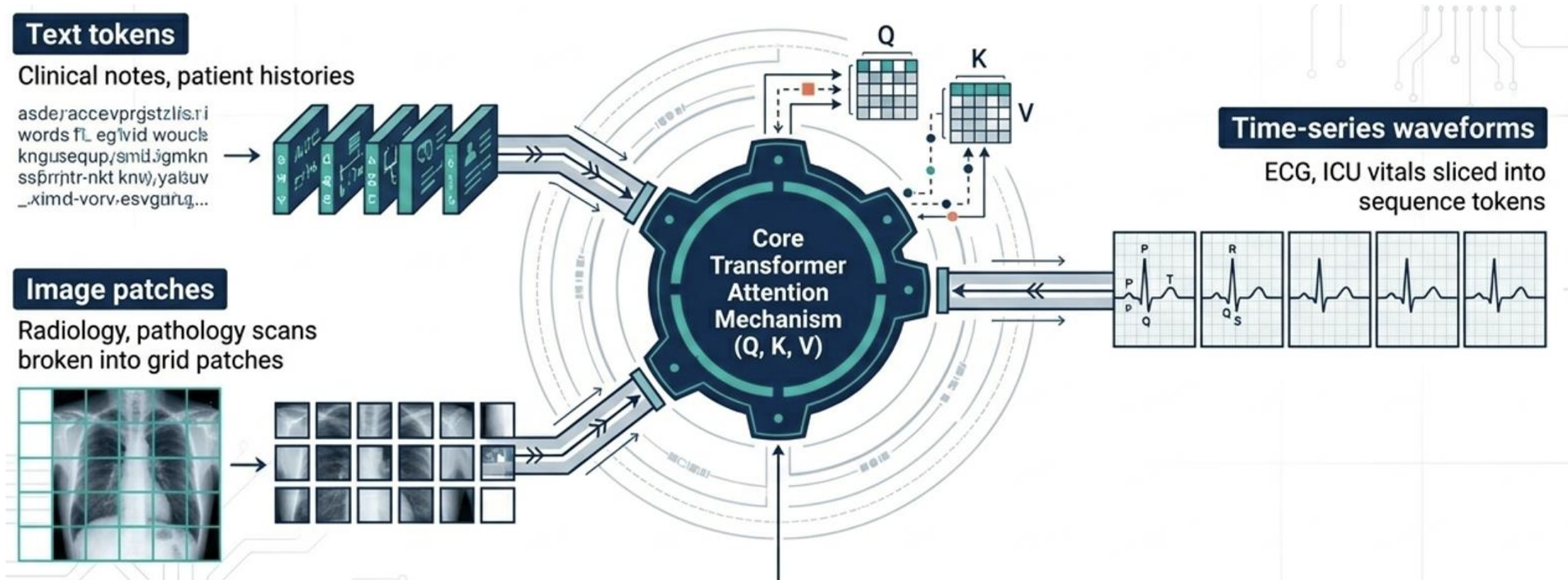
Instruct tuning bridges the gap between capability and safety

Evaluation Metric	Flan-PaLM (Unaligned)	Med-PaLM (Aligned)	Human Clinician
Scientific Consensus	61.9%	92.6%	92.9%
Incorrect Content	16.1%	18.7%	1.4%
Extent of Harm	29.7% (High Risk)	5.9% (Safe)	5.7% (Safe)
Evidence of Bias	7.9%	0.8%	1.4%

- Flan-PaLM is an instruct tuned LLM without clinical alignment
- Med-PaLM is fine-tuned by MultiMedQA
- Scientific consensus: from 61.9% to 92.6%
- Potential harm: from 29.7% to 5.9%, comparable to human expert
- Bias: from 7.9% to 0.8%, better than human expert

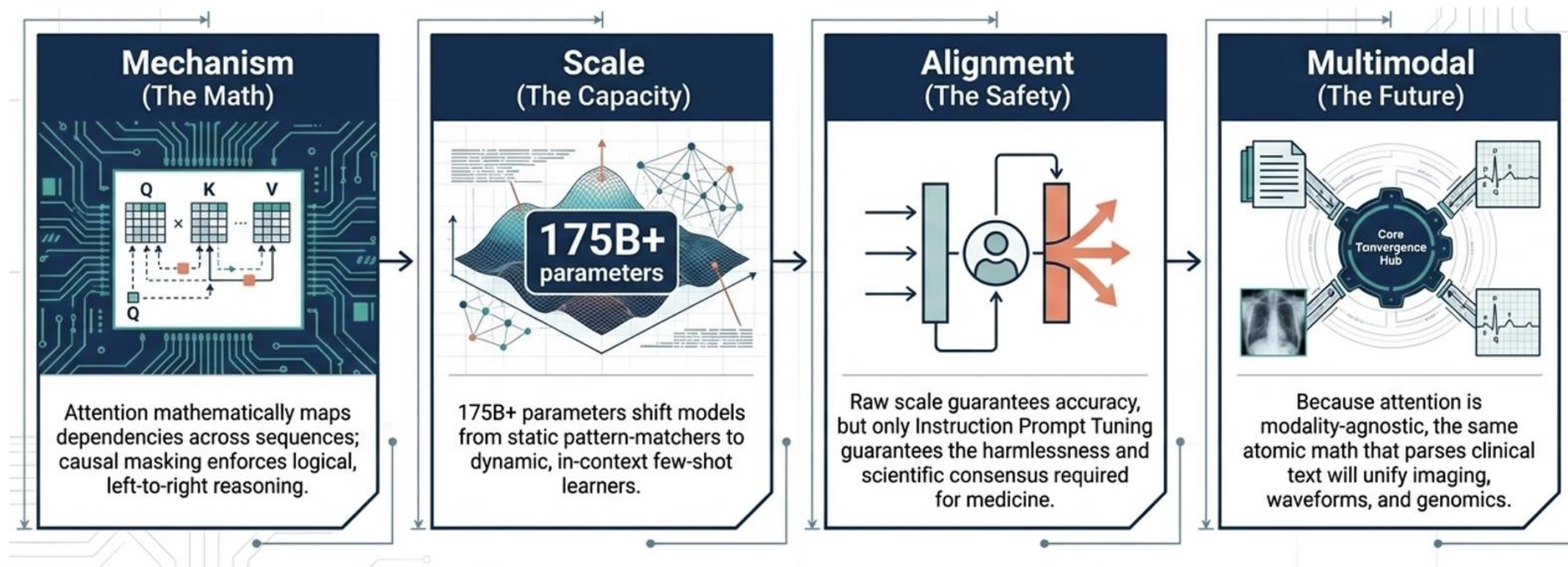
Multimodal roadmap

Universal architecture: convert knowledge into tokens



- Vision transformer (ViT) borrows the attention idea from GPT.
- Image is broken into patches and treated as sequence, then cross-referenced to text tokens
- Projecting image patches, temporal signals, clinical text into a shared latent space
- Same attention mechanism is driving multimodal AI

Blueprint of multimodal LLM



A synthesis of attention, scale, and sequence

Critical questions and challenges

Mechanism & Efficiency



Is self-attention's $O(n^2)$ computational cost acceptable for processing longitudinal patient records that span decades of history?

Scale vs. Specialization



For highly specialized medical research, when is in-context learning sufficient, and when is rigorous fine-tuning on proprietary data required?

Safety & Human Interaction



How should an AI system explicitly communicate its uncertainty, and what are the strict clinical thresholds for deferring to a human physician?